

Sarah L Kiefer

Department of Cognitive & Psychological Sciences, Brown University

sarah_kiefer@brown.edu | <https://slkiefer.github.io/>

EDUCATION

Ph.D. in Psychology

Expected May 2027

Department of Cognitive & Psychological Sciences, Brown University, Providence, RI

Proposed Dissertation: Children's understanding of productive failure in learning and persistence

Committee: Dr. David Sobel (Chair), Dr. Daphna Buchsbaum, Dr. Julia Leonard

Sc. M. in Cognitive Science

May 2024

Department of Cognitive, Linguistic & Psychological Sciences, Brown University, Providence, RI

Thesis: "I Wanted to Challenge Myself!": Children's Developing Metacognitive Understanding of Effort in a Building Task

Committee: Dr. David Sobel (Chair), Dr. Daphna Buchsbaum, Dr. Amitai Shenhav

Bachelor of Science, Psychology (Psychological Science)

May 2019

College of Liberal Arts and Sciences, Arizona State University, Tempe, AZ

HONORS AND AWARDS

Caplan Foundation for Early Childhood. Project title: Children's understanding of productive failure in learning and persistence. \$50,000 in total costs. Role: PI (Co-PI: David Sobel), 2026-2027

John Templeton Foundation Sub-award from the Developing Intellectual Humility Group, Humble beginnings: Exploring the role of intellectual humility in children's developing understanding of productive failure. \$38,733 in total costs. Role: Co-PI (PI: Dave Sobel), 2026-2027

Peter D. Eimas Graduate Research Award, Brown's Department of Cognitive and Psychological Sciences, 2025-2026.

NSF Graduate Research Fellowship Program Honorable Mention, 2024.

Graduate Conference Travel Award, Brown University. 2024, 2025, 2026.

PUBLICATIONS

*MENTORED STUDENT

Peer-Reviewed

Kiefer, S. L., & Sobel, D. M. (2026). Children's decision to challenge themselves on a novel task relates to their metacognitive monitoring of their ability. *Child Development*.

<https://doi.org/10.1093/chidev/aacaf045>

Vaisarova, J., Kiefer, S. L., Şen, H., Todd, P. M., & Lucca, K. (2024). Where should I search next? Messages embedded in storybooks influence children's strategic exploration in Turkey and the United States. *Journal of Experimental Psychology: General*, 153(8), 2174–2192.

<https://doi.org/10.1037/xge0001619>

Şen, H. H., Kiefer, S. L., Aksu, E., Lucca, K. (2024). Developmental differences in children and adults'

enforcement of explore versus exploit search strategies in the United States and Turkey.

Developmental Science, e13520. <https://doi.org/10.1111/desc.13520>

Kiefer, S. L., Caballero, A.*, Lucca, K. (2023). The role of effort type and intensity in children's decisions about effort-based outcomes. *Infant and Child Development*.

<http://doi.org/10.1002/icd.2440>

Matthews, N. L., Smith, C. J., Skepnek, E., Mammen, M., James, J. S., Malligo, A., Lyon, A., Mitchell, M.,

Kiefer, S. L., Smith, C. J. (2021). Feasibility and acceptability of a telehealth model for autism

diagnostic evaluations in children, adolescents, and adults. *Autism Research*. DOI: 10.1002/aur.2591

Matthews, N. L., Christenson, K., **Kiefer, S.,** Smith, C. J. (2019). A mixed-methods examination of

the gap between IQ and adaptive functioning in autistic young adults without

intellectual disability. *Autism*. DOI: 10.1177/13623613211018334

Manuscripts in Preparation and Under Review

Kiefer, S. L., Modesti, G., McHugh, S., & Sobel, D. M. (submitted for revised review, *Journal of Cognition and Development*). *The STEM Engine: Fostering early engagement with STEM reasoning challenges through parent-child interaction*. Brown University.

McHugh, S., Modesti, G., **Kiefer, S. L.,** & Sobel, D. M. (in progress). Parent-child interaction and children's learning and engagement with engineering principles. Brown University.

Kiefer, S. L., & Sobel, D.M. (in preparation). *The role of productive failure pedagogy in children's hypothesis testing*. Brown University.

Kiefer, S. L., & Sobel, D.M. (in preparation). *Children's developing integration of accuracy and difficulty in understanding practice*. Brown University.

Kiefer, S. L., & Sobel, D.M. (in preparation). *Children's developing understanding of strategic practice*. Brown University.

Kiefer, S. L. (in preparation). *Towards an integrated developmental account of persistence*. Brown University.

CONFERENCE PRESENTATIONS

*MENTORED STUDENT

Chaired Symposia

Mahaffey, E., **Kiefer, S. L.** (2025, May). The effort equation: Towards an integrative understanding of the factors that drive children's adaptive persistence. *Society for Research in Child Development*. Minneapolis, MN, USA.

Oral Presentations

Kiefer, S. L., Sobel, D. M. (2025, May). Children's decisions about difficulty during novel tasks relate to their metacognitive monitoring of their ability. *Talk presented at the Society for Research in Child Development*. Minneapolis, MN, USA.

Kiefer, S. L., Caballero, A.*, Lucca, K. (2022, November). The role of effort type and intensity in children's decision making. *Flash talk at the 3rd annual workshop on mental effort*. Providence, RI, USA.

Poster Presentations

- Kiefer, S. L.,** Modesti, G., McHugh, S., Sobel, D. M. (2026, April) Fostering STEM Engagement in Spatial Reasoning from Parent-Child Interaction: A Remote Study. *Poster presented at the Cognitive Development Society Biennial Meeting*. Montreal, Quebec, Canada.
- Kiefer, S. L.,** Vezina, O.*, Sobel, D. M. (2025, May). “That was easy!” Children’s definitions and examples of easy and hard experiences. *Poster presented at the Society for Research in Child Development*. Minneapolis, MN, USA.
- Kiefer, S. L.,** Sobel, D. M. (2024, March). “I wanted to challenge myself!”: Children’s developing metacognitive understanding of effort allocation in a building task. *Poster presentation at the Cognitive Development Society Biennial Meeting*. Pasadena, CA, USA.
- Vaisarova, J. (presenting author), **Kiefer, S. L.,** Şen, H. H., Todd, P., & Lucca, K. (2024, March). Where Should I Look Next? Using a Storybook Intervention to Promote Children’s Strategic Exploration in a Virtual Search Task Across Cultures. *Poster presentation at the Cognitive Development Society Biennial Meeting*. Pasadena, CA, USA.
- Özalp, D., **Kiefer, S. L.,** Uğur, G., Hanım Sancar, B., Ateş, S., Selalmaz, E., Uçal, A., Nur Deveci, A., Akpınar, B., Karaoğlu, M., Şen, H. (presenting author), Lucca, K. (2023, January). Associations between mothers’ and children’s curiosity in the Turkish context. *Poster Presentation at the Budapest CEU Conference on Cognitive Development*. Budapest, Hungary.
- Kiefer, S. L.,** Caballero, A.*, Lucca, K. (2022, November). The role of effort type and intensity in children’s decision making. *Poster presentation at the 3rd annual workshop on mental effort*. Providence, RI, USA.
- Johns, A. (presenting author), Anbar, J., **Kiefer, S. L.,** Smith, C. J. (2022, May). Building inclusive informal STEM learning opportunities for people with ASD. *International Society for Autism Research Annual Meeting*. Austin, TX, USA.
- Kiefer, S. L.,** Şen, H., Akpınar, B., Smith, B.*, Nur Deveci, A., Selalmaz, E., Snodgrass, N.*, Özalp, D., Patel, S.*, Uçal, A., Lucca, K. (2022, April). Navigating the explore-exploit dilemma: a cross-cultural examination of 3–6-year-olds’ own behavioral strategies and enforcement of strategies onto others. *Poster presentation at the Cognitive Development Society Biennial Meeting*. Madison, WI, USA.
- Kiefer, S. L.,** Caballero, A.*, Lucca, K. (2022, April). Worth the wait: the role of cognitive and physical effort in shaping children’s value judgments, preferences, and behavior. *Poster presentation at the Cognitive Development Society Biennial Meeting*. Madison, WI, USA.
- Saguid, L. (presenting author), Johnson, N.*, **Kiefer, S.,** Scirpo, K.*, Patel, S.*, Hinojosa, A.*, Snodgrass, N.*, Kupfer, A., Lucca, K. (2021, September). Investigating the relations between early creativity, confidence, curiosity, and persistence. *Poster presentation at the Marconi Institute for Creativity Conference*. Online.
- Christenson, K., **Kiefer, S.,** Tracey, S., Matthews, N. L., & Smith, C. J. (2019, May). A qualitative examination of the transportation experiences and needs of young adults with ASD without intellectual disability. *Poster presentation at the International Society for Autism Research Annual Meeting*. Montreal, Quebec, Canada.

INVITED TALKS

October, 2026 Social Learning Lab, Stanford University

October, 2026	Social Learning Lab, Boston University
August, 2026	Centre for Developing Minds, Trent University
July, 2026	Leonard Learning Lab, Yale University
June, 2026	Early Childhood Cognition Lab, Duke University
May, 2026	Early Learning and Cognition Lab, University of California, San Diego
November, 2025	Developmental Brown Bag, Brown University
January, 2022	Cognition and Learning Center, Rutgers University

MEDIA COVERAGE

Rajpal, A. (2026, March 11). Children seek further challenges when they succeed on difficult tasks, Brown study finds. *Brown Daily Herald*. Available at <https://www.browndailyherald.com/article/2026/03/children-seek-further-challenges-when-they-succeed-on-difficult-tasks-brown-study-finds>

Abrams Gerber, H. (2025, January 27). Playing with their parents: children and the development of scientific reasoning. *COPSY News*. Available at <https://copsy.brown.edu/news/2026-01-27/new-research-professor-dave-sobel-studies-how-we-learn-learn>

Stanford Psychology Podcast. (2024, July 25). 137 – Kelsey Lucca: Unpacking the development of exploration and exploitation [Podcast episode]. In Stanford Psychology Podcast. Available at <https://www.stanfordpsychologypodcast.com/episodes/episode/807a22a2/137-kelsey-lucca-unpacking-the-development-of-exploration-and-exploitation>

Ewing, R. (2022, August 15). Emerging Minds Lab receives NSF CAREER grant to expand access to research. *ASU News*. Available at <https://news.asu.edu/20220815-emerging-minds-lab-receives-nsf-career-grant-expand-access-research>

Greguska, E. (2021, April 6). Tiny volunteers help ASU lab study curiosity. *ASU News*. Available at <https://news.asu.edu/20210405-solutions-baby-einsteins>

PRE-DOCTORAL RESEARCH EXPERIENCE

Emerging Minds Lab | PI: Dr. Kelsey Lucca | Tempe, AZ

June 2021 – June 2022 | **Lab Manager**

Feb. 2020 – June 2021 | **Project Coordinator**

Oct. 2019 – Feb. 2020 | **Research Assistant**

Southwest Autism Research and Resource Center (SARRC) | Phoenix, AZ

Jan. 2021 – May 2021 | **Research Study Coordinator**

Dec. 2018 – Dec. 2020 | **Research Study Recruiter**

May 2018 – May 2019 | **Research Intern**

REACH Institute, Bridges to High School Program | PI: Dr. Nancy Gonzales | Tempe, AZ

Jan. 2018 – May 2019 | **Data Management Research Assistant & Phone Interviewer**

Aug 2017 – Dec. 2017 | **In-home Interviewer**

TEACHING EXPERIENCE

Course Instructor

Summer 2026 Emerging Minds: How children learn to think and think to learn, Summer@Brown pre-college program

Teaching Assistant, Brown University

Spring 2026 Children's Thinking: The Nature of Cognitive Development

Fall 2025 Social and Moral Development

Spring 2025 Children's Thinking: The Nature of Cognitive Development

Summer 2024 Psychology Across Species, Summer@Brown pre-college program

Spring 2024 Statistical Methods

Guest Lectures

Jan. 2026 Children's Thinking: The Nature of Cognitive Development
Brown University, Dr. Daphna Buchsbaum

Feb. 2025 Children's Thinking: The Nature of Cognitive Development
Brown University, Dr. David Sobel

Sept. 2021 Cognitive Development
Arizona State University, Dr. Kelsey Lucca

Feb. 2020 Cognitive Development
Arizona State University, Dr. Kelsey Lucca

Select Professional Development

Brown University Sheridan Center for Teaching and Learning: Teaching Consultant Program, Fall 2026

Brown University Sheridan Center for Teaching and Learning: Sheridan Teaching Seminar Certificate, Fall 2024

Center for the Integration of Research, Teaching, and Learning (CIRTL):

- Pedagogies of Care Workshop, January 21st, 2026
- Decentering Grades: Getting Started with Ungrading, March 2nd, 2026
- The Joyful Journey: Teaching, Research, and Diversity Statement Workshops (January – March, 2026)

MENTORING EXPERIENCE

Undergraduate Teaching and Research Award Brown University, Students Supervised:

- Lorine Makokha (Fall 2026)
- Similolu Fadel (Spring 2026)
- Samaira Gupta (Spring 2026)
- Katherine Kupsaw (Spring 2024)
- Seidy Idris (Spring 2024)

Barrett Honors College Thesis Funding Award, Arizona State University, Students Supervised:

- Nicole Johnson, 2021-2022
- Kaitlin Van Houghton, 2021-2022
- Sonia Patel, 2021-2022
- Kalie Scirpo, 2020-2021

PROFESSIONAL SKILLS

Programming: Proficiency in R, Jamovi, JASP, SPSS. Some proficiency in JSON (i.e., programming experiments on childrenhelpingscience.com)

Computer/Web Applications: Adobe Suite (Illustrator, Photoshop), Squarespace, Datavyu, Salesforce, REDCap, Qualtrics, MTurk

Administration of behavioral experiments with children: over 600 participants tested to date.

Assessments: Experienced in administering cognitive and behavioral assessments across the lifespan, including the Mullen Scales of Early Learning and the Kaufman Brief Intelligence Test. Received extensive training on the Autism Diagnostic Observation Schedule, Autism Diagnostic Interview-Revised, Preschool Language Scales, and Wechsler Intelligence Scale for Children (full certification not complete due to COVID-19 interruptions).

ACADEMIC SERVICE

Spring 2024 – Present	Diversity and Inclusion Action Plan Committee, Community Outreach Subcommittee, Department of Cognitive and Psychological Sciences at Brown University
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Fall 2023 – Spring 2024	Graduate Student Council Representative for the Department of Cognitive, Linguistic & Psychological Sciences at Brown University
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PROFESSIONAL AFFILIATIONS AND WORKSHOPS

Spring 2025 - Present	Member of the Society for Research in Child Development
Spring 2022 - Present	Member of the Cognitive Development Society
Summer 2023	Computational Cognitive Models of Learning and Development Workshop – Harvard University
Fall 2021 and Fall 2022	Workshop on Mental Effort
Spring 2019 – Spring 2020	Member of the International Society for Autism Research

COMMUNITY ENGAGEMENT

Spring 2023, 2024	<u>College Day at Brown</u> facilitator for the Department of Cognitive & Psychological Sciences
Fall 2026 – Present	Visiting Field Researcher, Southwick's Zoo
Fall 2022 – Fall 2025	Visiting Field Researcher, Roger Williams Park Zoo
Fall 2022 – Fall 2023	Mind Lab Researcher, Providence Children's Museum
Spring 2023 – Fall 2024	In collaboration with Dr. Amitai Shenhav, facilitated motivation and effort lessons for 4 th graders at William D'Abate Elementary in Providence. We visited weekly across four weeks with two cohorts of 4 th graders.